

A Strictly Convex c_0 Half-Space With No Quasihyperbolic Geodesics

Run `fa.banach.001`, target 1301.0900

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Source and Open Problem

The source is Rasila–Talponen, *On Quasihyperbolic Geodesics in Banach Spaces*, arXiv:1301.0900 [1]. The target signal is Conjecture 2.5 on page 7:

We conjecture that a strictly convex Banach space is reflexive if and only if each of its open half spaces is geodesic in the quasihyperbolic metric.

The same page notes, without proof, that the preceding c_0 half-space example can be modified so that the ambient Banach space is strictly convex. The crop below records both the conjecture and that omitted-proof remark.

This means that $\ell_{\text{qh}}(\gamma_k) < \ell_{\text{qh}}(\gamma)$ for k large enough. This provides us with a contradiction, since γ was assumed to be a geodesic. $\square$

Suppose that X is a Banach space with $\dim X \geq 2$, and let $\Omega \subsetneq X$ be an open path-connected domain. Recall that a domain Ω is geodesic if for every pair of points $x, y \in \Omega$ there exists a geodesic connecting x and y .

Conjecture 2.5. We conjecture that a strictly convex Banach space is reflexive if and only if each of its open half spaces is geodesic in the quasihyperbolic metric. As it was established above, the 'only if' direction holds (since reflexivity implies the RNP).

In the non-strictly convex case the geodesics of half spaces need not be unique, even in finite dimensional setting. For example, let $\Omega = \{(x, y) \in \mathbb{R}^2 : y > 0\} \subset c_{00}(2)$ (with the sup-norm). Then the straight line between $(0, 1)$ and $(0, 2)$ is a quasihyperbolic geodesic. Note that the polygonal line between these two points passing through $(1/2, 3/2)$ is also a quasihyperbolic geodesic.

Without giving a proof we note that the above half-space example on c_0 can be modified in such a way that the Banach space can be taken to be even strictly convex. Indeed, a well-known equivalent strictly convex norm $||| \cdot |||$ on c_0 is given by

$$|||(x_n)|||^2 = \|(x_n)\|_{c_0}^2 + \sum_{n=1}^{\infty} 2^{-n} |x_n|^2$$

and $(f_n) \in \ell^1$, $\|(f_n)\| = 1$, can be selected in such a way that $\limsup_n f_n/2^{-n} = \infty$. Note that $|||(f_n)|||^* \leq 1$, so that $\text{dist}(x, \text{Ker } f) \geq \langle f, x \rangle$.

3. SMOOTHNESS OF GEODESICS

The modulus of convexity δ_X of Banach space is not necessarily convex. However, it has a largest convex minorant δ . It is easy to see that this is strictly positive if δ_X is such. In what follows we will use the greatest convex minorant δ in the place of δ_X . A modulus of convexity δ_X is said to have a power type if there is $p \geq 2$ such that $\delta_X(\epsilon) \geq c\epsilon^p$.

Result Status

This packet proves the strict-convex c_0 subcase announced in the source remark. It is a partial result relative to Conjecture 2.5, not a proof of the full conjecture. In particular, the argument uses the coordinate structure of c_0 and does not show that every nonreflexive strictly convex Banach space has a non-geodesic half-space.

Definitions

For a domain Ω in a Banach space X , the quasihyperbolic length of a rectifiable curve γ is

$$\ell_k(\gamma) = \int \frac{\|d\gamma\|}{\text{dist}(\gamma, \partial\Omega)}.$$

The domain is geodesic if every two points in it can be joined by a curve attaining the corresponding infimum.

Let $w_n = 2^{-n}$. On c_0 define

$$\mathcal{N}(x)^2 = \|x\|_\infty^2 + \sum_{n=1}^{\infty} w_n |x_n|^2.$$

This norm is equivalent to the usual c_0 -norm and is strictly convex, because the positive quadratic summand is strictly convex and is added to a convex function.

Proof Intuition

The original c_0 proof with the sup norm raises many far coordinates of an alleged geodesic. Raising coordinates does not increase the sup-norm speed, but it increases the functional f , hence increases the distance to the boundary of the half-space and lowers quasihyperbolic length.

The strict convex perturbation adds a weighted Hilbert term. A single far-coordinate bump can increase this norm only by a quantity proportional to 2^{-m} , while it increases the denominator by a quantity proportional to f_m . Taking m with $f_m/2^{-m}$ very large makes the denominator improvement dominate. Thus every alleged geodesic can be shortened.

Lemma 1 (one-coordinate speed estimate). *Let $\gamma : [0, 1] \rightarrow c_0$ be Lipschitz for $\|\cdot\|_\infty$, parametrized with constant sup-metric speed $S > 0$. Fix m . Suppose Γ is obtained from γ by changing only the m -th coordinate, and the new m -th coordinate is also S -Lipschitz. Then, for almost every t ,*

$$\text{md}_{\mathcal{N}} \Gamma(t) \leq \text{md}_{\mathcal{N}} \gamma(t) + Sw_m.$$

Proof. The metric derivative exists almost everywhere for Lipschitz curves. At almost every t , the scalar derivatives of the relevant coordinate functions also exist. Since γ has constant sup-metric speed S , the modified curve has sup-metric derivative at most S . All weighted square-coordinate contributions agree with those of γ , except possibly the m -th one, where the new scalar derivative has absolute value at most S . Hence

$$\text{md}_{\mathcal{N}} \Gamma(t)^2 \leq \text{md}_{\mathcal{N}} \gamma(t)^2 + w_m S^2.$$

Also $\text{md}_{\mathcal{N}} \gamma(t) \geq \text{md}_\infty \gamma(t) = S$ almost everywhere. Therefore

$$\text{md}_{\mathcal{N}} \Gamma(t) \leq \sqrt{\text{md}_{\mathcal{N}} \gamma(t)^2 + w_m S^2} \leq \text{md}_{\mathcal{N}} \gamma(t) + Sw_m,$$

using $0 < w_m \leq 1$. □

Theorem 1. *There are an equivalent strictly convex norm $\mathcal{N}$ on c_0 and a continuous linear functional $f \in (c_0, \mathcal{N})^*$ such that the open half-space*

$$\Omega = \{x \in c_0 : f(x) > 0\}$$

is not geodesic in the quasihyperbolic metric induced by $\mathcal{N}$. More precisely, no two distinct points of Ω are joined by a quasihyperbolic geodesic.

Proof. Let

$$f(x) = \sum_{n=1}^{\infty} f_n x_n,$$

where $f_n > 0$, $\sum_n f_n = 1$, and f_n/w_n is unbounded. For instance, after normalization one may take $f_n = n^{-2}$. Since $\mathcal{N}(x) \geq \|x\|_{\infty}$, the functional f is continuous. For $x \in \Omega$,

$$\text{dist}_{\mathcal{N}}(x, \partial\Omega) = \frac{f(x)}{\|f\|_{\mathcal{N},*}},$$

so the constant factor $\|f\|_{\mathcal{N},*}$ is irrelevant for comparing quasihyperbolic lengths. It is enough to compare

$$L_f(\sigma) = \int_0^1 \frac{\text{md}_{\mathcal{N}} \sigma(t)}{f(\sigma(t))} dt.$$

Assume toward a contradiction that a quasihyperbolic geodesic γ joins distinct points $x, y \in \Omega$. Reparametrize γ on $[0, 1]$ by constant $\|\cdot\|_{\infty}$ -arclength speed $S > 0$. The image of γ is compact in Ω , so

$$a := \min_t f(\gamma(t)) > 0, \quad A := \max_t f(\gamma(t)) < \infty,$$

and

$$C := \int_0^1 \frac{dt}{f(\gamma(t))} < \infty.$$

Compactness of $\gamma([0, 1]) \subset c_0$ gives uniform tail decay. Choose m so large that

$$\sup_{t \in [0, 1]} |\gamma_m(t)| < S/16, \quad |x_m| < S/16, \quad |y_m| < S/16,$$

and also

$$\frac{f_m}{w_m} > \frac{16(A + S)^2 C}{S}.$$

Set

$$\alpha(t) = \min(x_m, y_m) + S \min(t, 1 - t)$$

and define Γ by

$$\Gamma_n(t) = \gamma_n(t) \quad (n \neq m), \quad \Gamma_m(t) = \max\{\gamma_m(t), \alpha(t)\}.$$

Then $\Gamma(0) = x$, $\Gamma(1) = y$, and $\Gamma(t) \in c_0$. Moreover $q(t) := \Gamma_m(t) - \gamma_m(t) \geq 0$, so $\Gamma(t) \in \Omega$. On $I = [1/4, 3/4]$, the tail choice yields

$$q(t) \geq S/8.$$

Also $q(t) \leq S$ for all t , after increasing m if necessary within the same tail condition.

By Lemma 1,

$$L_f(\Gamma) - L_f(\gamma) \leq \int_0^1 \text{md}_{\mathcal{N}} \gamma(t) \left(\frac{1}{f(\gamma(t)) + f_m q(t)} - \frac{1}{f(\gamma(t))} \right) dt + S w_m C.$$

The first integrand is nonpositive everywhere. On I , using $\text{md}_{\mathcal{N}} \gamma(t) \geq S$, $q(t) \geq S/8$, and $f(\gamma(t)) + f_m q(t) \leq A + S$, it is at most

$$-\frac{S \cdot f_m(S/8)}{(A + S)^2}.$$

Since $|I| = 1/2$, we get

$$L_f(\Gamma) - L_f(\gamma) \leq Sw_m C - \frac{f_m S^2}{16(A + S)^2} < 0$$

by the choice of m . Thus Γ has strictly smaller quasihyperbolic length than γ , contradicting the assumption that γ was geodesic.

The endpoints $x \neq y$ were arbitrary. Hence no distinct pair of points in Ω is joined by a quasihyperbolic geodesic. $\square$

Verification Notes

The proof uses no numerical search. The only analytic inputs beyond the source definitions are standard facts about metric derivatives of Lipschitz curves and the elementary formula $\text{dist}(x, \ker f) = |f(x)|/\|f\|$ for a continuous nonzero linear functional.

The key inequality to review is Lemma 1; it is the place where the strict convex perturbation is controlled. The argument changes a single far coordinate, so the extra Hilbert-square contribution is $w_m S^2$, while the half-space denominator gain is $f_m q(t)$ on a fixed middle interval.

Limitations and Novelty Check

This is not a full solution of Conjecture 2.5. The proof depends on the coordinate lattice structure of c_0 and on choosing $f_m/2^{-m}$ unbounded. It does not address arbitrary nonreflexive strictly convex Banach spaces.

Novelty confidence is intentionally modest: the source paper itself states the strict-convex modification without proof. The contribution of this packet is to make that omitted argument explicit and reviewable. I searched the run indexes for 1301.0900, quasihyperbolic half-spaces, and the strict-convex/reflexive keywords, and checked later quasihyperbolic follow-up material including Rasila–Talponen–Zhang [2]. I found no later local packet or obvious later paper settling the full conjecture; the 2017 follow-up still records a broader reflexivity-converse problem.

Human Review Recommendation

Review as a partial packet. The result is compatible with, but does not settle, Rasila–Talponen Conjecture 2.5. The main review focus should be the metric-derivative estimate under the one-coordinate max modification and the strict inequality in the final length comparison.

References

- [1] A. Rasila and J. Talponen, *On Quasihyperbolic Geodesics in Banach Spaces*, Ann. Acad. Sci. Fenn. Math. 39 (2014), 163–173. arXiv:1301.0900.
- [2] A. Rasila, J. Talponen, and X. Zhang, *Observations on quasihyperbolic geometry modeled on Banach spaces*, arXiv:1708.02240.